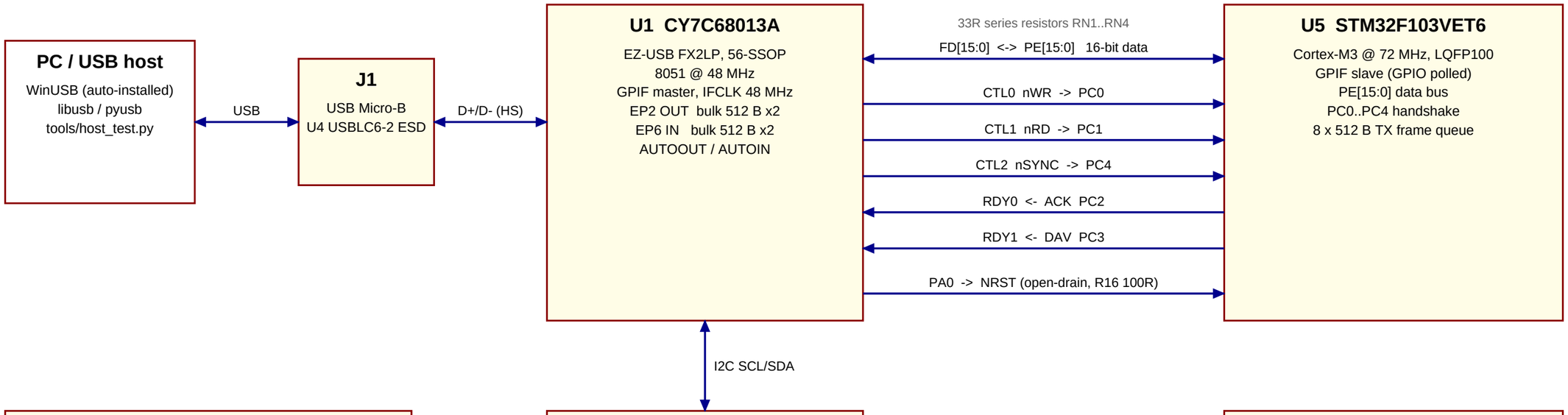
FX2LP (CY7C68013A-56PVXC) <-> STM32F103VET6 : USB 2.0 High-Speed bridge over a 16-bit GPIF bus



GPIF interconnect

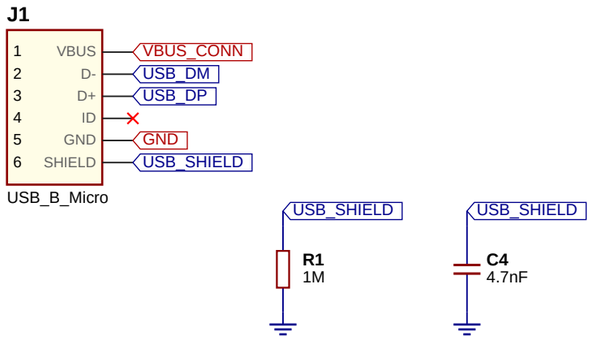
Signal	FX2 pin (56-SSOP)	Net	STM32 pin (LQFP100)	Dir	Function
FD0..FD7	PB0..PB7 : 25..32	FD0..7 / GD0..7	PE0..PE7 : 97,98,1,2,3,4,5,38	<->	data bus low byte (1st byte of USB data)
FD8..FD15	PD0..PD7 : 52..56,1,2,3	FD8..15 / GD8..15	PE8..PE15 : 39..46	<->	data bus high byte
nWR	CTL0 : 36	GPIF_NWR	PC0 : 15	->	FX2 write strobe, low = data valid
nRD	CTL1 : 37	GPIF_NRD	PC1 : 16	->	FX2 read strobe, low = STM32 must drive bus
nSYNC	CTL2 : 38	GPIF_NSYNC	PC4 : 33	->	low pulse = drop partial frames (FIFO reset)
ACK	RDY0 : 8	GPIF_ACK	PC2 : 17	<-	word handshake (4-phase)
DAV	RDY1 : 9	GPIF_DAV	PC3 : 18	<-	STM32 has a 512-byte frame for the PC
NRST	PA0 : 40	STM_NRST_CTL	NRST : 14	->	FX2 can reset the STM32 (vendor req. 0xB1)

Design notes

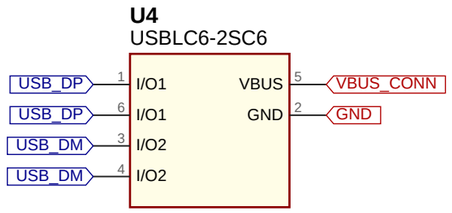
1. One GPIF transaction = one 16-bit word, fully handshaked:
   the STM32 serves the bus by GPIO polling, no timing constraint.
2. USB data never passes through the 8051 (AUTOOUT/AUTOIN FIFOs).
3. All transfers are 512-byte frames (one HS packet).
4. VID/PID 04B4:F103 is for development only - use your own.
5. Decouple every VCC/VDD pin with 100 nF placed at the pin.
6. USB D+/D-: 90 ohm differential, short, no stubs, no vias
   if possible; keep the 24 MHz crystal next to U1.
7. Keep FD/CTL/RDY traces < 10 cm, series resistors at U1.
8. Unused FX2 PA2..PA7 are driven low by firmware.

PROJECT	FX2LP - STM32F103 GPIF USB Bridge	
SHEET TITLE	Overview & GPIF interface	
DATE	REV	SHEET
2026-10-02	1.0	1 of 5

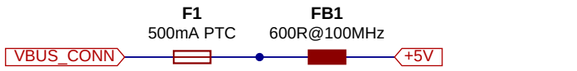
USB 2.0 connector



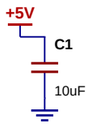
ESD protection (place next to J1)

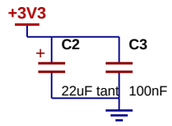


VBUS input



3.3 V regulator (AMS1117-3.3, 1 A)

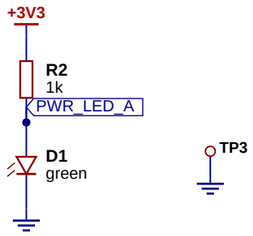




C1: VBUS bulk must stay <= 10 uF (USB 2.0 inrush limit)

C2: AMS1117 needs >= 22 uF tantalum for stability

Power LED

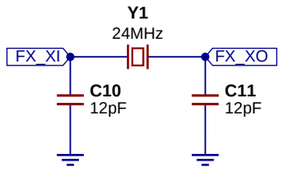


Notes

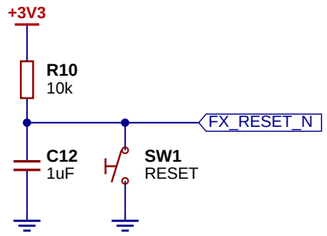
Board is bus powered: FX2LP <= 85 mA, STM32 <= 50 mA, LEDs ~ 8 mA
-> bMaxPower = 250 mA in the USB configuration descriptor.
USB_SHIELD is tied to GND through 1M || 4.7 nF near J1.
U4 must sit between J1 and U1 with straight-through D+/D- routing.

PROJECT		
FX2LP - STM32F103 GPIF USB Bridge		
SHEET TITLE		
USB connector & power supply		
DATE	REV	SHEET
2026-10-02	1.0	2 of 5

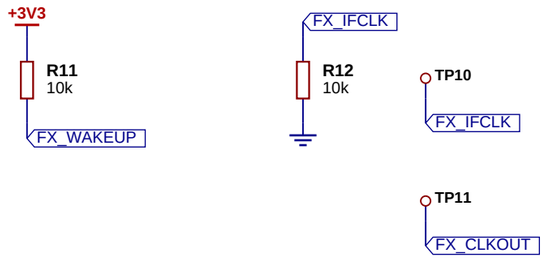
24 MHz crystal



Reset (> 5 ms after VCC)

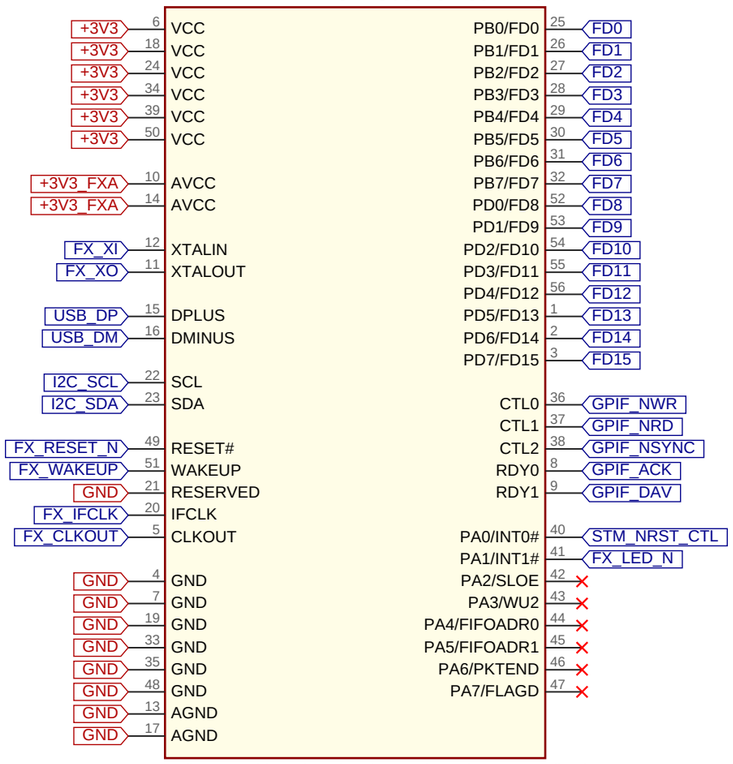


Misc. pins



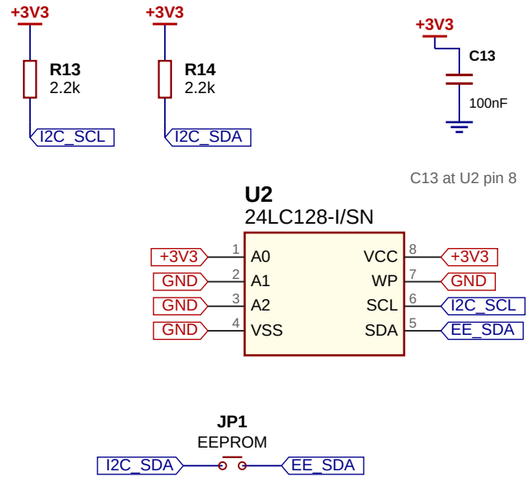
IFCLK pin is not driven (IFCONFIG.5 = 0), R12 keeps it defined.

U1 CY7C68013A-56PVXC



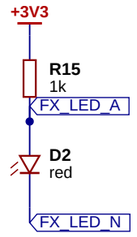
PA2..PA7: firmware drives them low (outputs), so they are not floating

Boot EEPROM (C2 load, address 0x51)

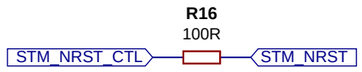


JP1 open: FX2 boots without EEPROM and enumerates as 04B4:8613 (first programming / recovery).

FX2 activity LED (PA1, active low)

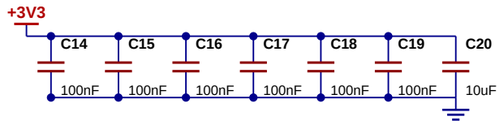


STM32 reset from FX2 (PA0)

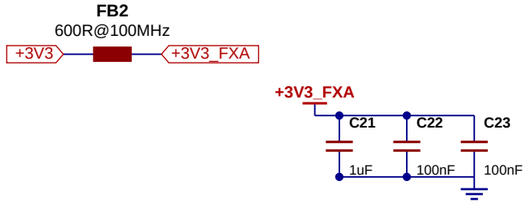


PA0 is only ever driven low (OEA.0 = 1 with IOA.0 = 0): open-drain emulation, no conflict with SW2 / ST-Link.

Decoupling (one 100 nF per VCC pin, at the pin)

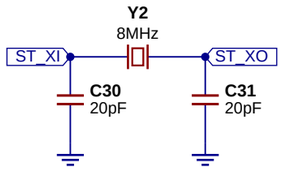


Analog supply AVCC

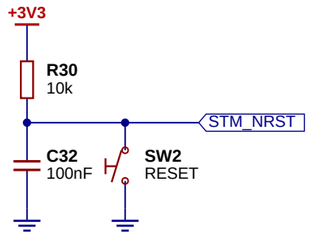


PROJECT		
FX2LP - STM32F103 GPIF USB Bridge		
SHEET TITLE		
CY7C68013A-56PVXC (EZ-USB FX2LP)		
DATE	REV	SHEET
2026-10-02	1.0	3 of 5

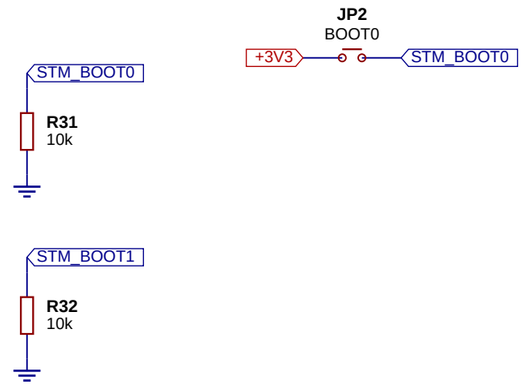
8 MHz crystal (HSE)



Reset



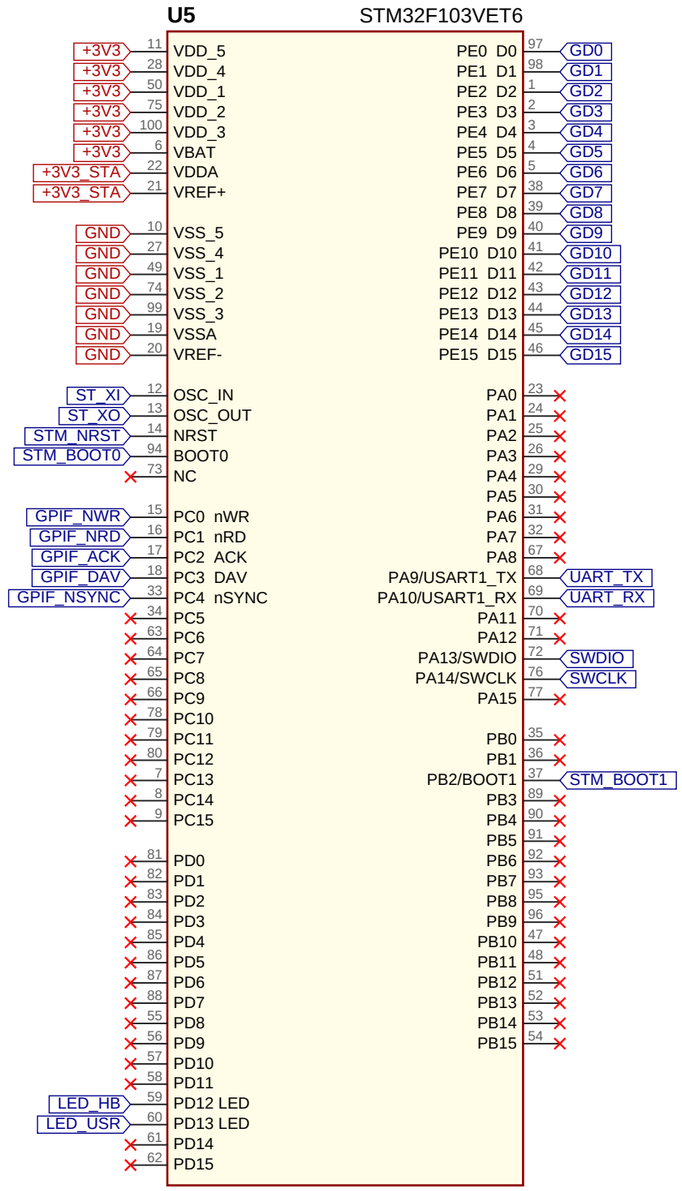
Boot mode (BOOT0 = 0: flash)



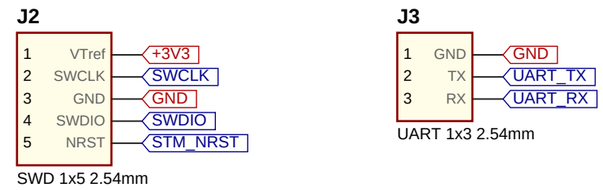
Notes

Unused GPIOs (X) stay in reset state (floating input);
firmware may set them to analog mode for lowest power.
PA13/PA14 remain SWD (default after reset).
PE[15:0] switch between input and output at run time;
the 33R series resistors (sheet 5) limit any contention.

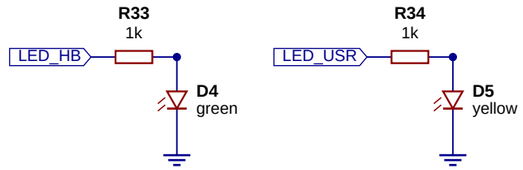
U5 STM32F103VET6



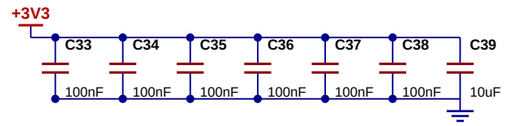
Debug connectors



LEDs (active high)

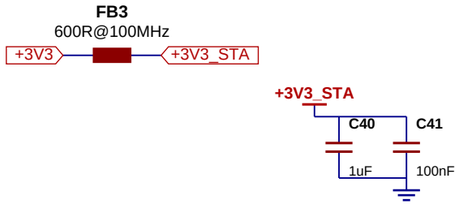


Decoupling (100 nF at each VDD pin)



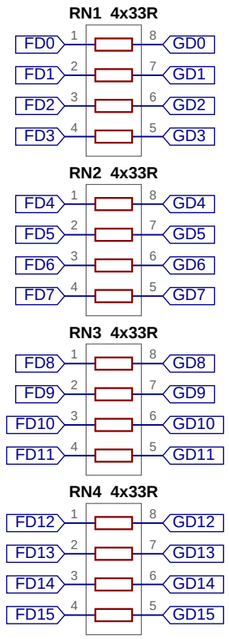
C38: VBAT

Analog supply VDDA / VREF+

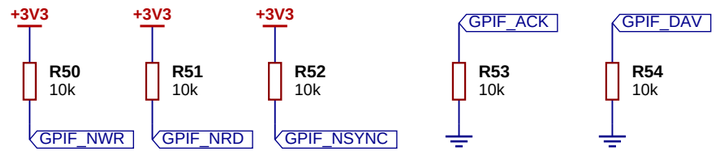


PROJECT	FX2LP - STM32F103 GPIF USB Bridge
SHEET TITLE	STM32F103VET6 (GPIF slave)
DATE	2026-10-02
REV	1.0
SHEET	4 of 5

Data bus series resistors (place close to U1)

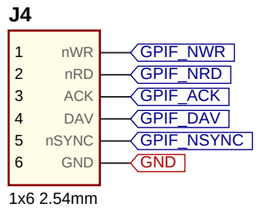


Handshake line bias (defined levels while a chip is in reset)



Pull-ups keep strobes inactive, pull-downs keep ACK/DAV = 0
(FX2 never sees 'ready' from an STM32 that is in reset).

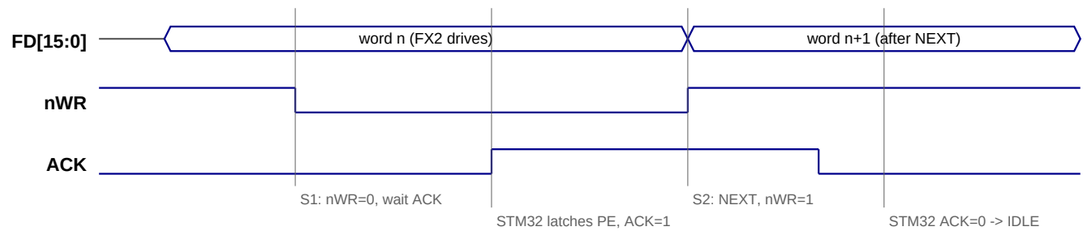
Logic analyser header



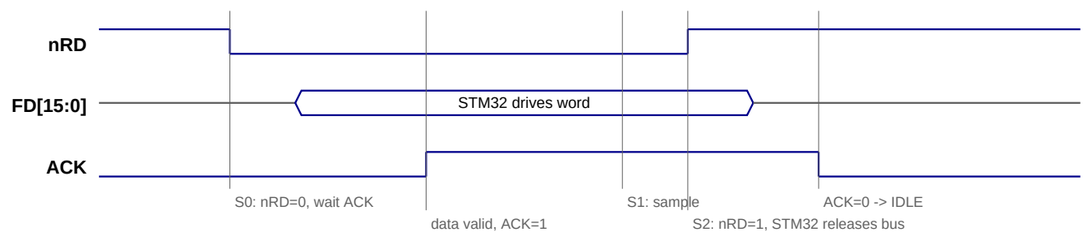
Bus mapping

FX2 pin	FX2 signal	net (FX2 / STM32)	STM32 pin
25	PB0 / FD0	FD0 / GD0	97 PE0
26	PB1 / FD1	FD1 / GD1	98 PE1
27	PB2 / FD2	FD2 / GD2	1 PE2
28	PB3 / FD3	FD3 / GD3	2 PE3
29	PB4 / FD4	FD4 / GD4	3 PE4
30	PB5 / FD5	FD5 / GD5	4 PE5
31	PB6 / FD6	FD6 / GD6	5 PE6
32	PB7 / FD7	FD7 / GD7	38 PE7
52	PD0 / FD8	FD8 / GD8	39 PE8
53	PD1 / FD9	FD9 / GD9	40 PE9
54	PD2 / FD10	FD10 / GD10	41 PE10
55	PD3 / FD11	FD11 / GD11	42 PE11
56	PD4 / FD12	FD12 / GD12	43 PE12
1	PD5 / FD13	FD13 / GD13	44 PE13
2	PD6 / FD14	FD14 / GD14	45 PE14
3	PD7 / FD15	FD15 / GD15	46 PE15

FX2 -> STM32 word write (GPIF FIFO-write waveform, 1 transaction)



STM32 -> FX2 word read (GPIF FIFO-read waveform, 1 transaction)



Protocol

DAV (RDY1) is high while the STM32 has a 512-byte frame queued.
FX2 reads one USB packet per GPIF trigger (256 words HS, 32 words FS)
only when DAV = 1 and the EP6 FIFO is not full.
FX2 writes one EP2 packet per trigger (byte count / 2 words).
nSYNC low (1 ms): issued after power-up, SET_CONFIGURATION and
vendor request 0xB1 - STM32 drops partial frames on both sides.
All RDY inputs are synchronised by the FX2 (IFCONFIG.ASYNC = 1),
so no setup/hold requirement exists on the STM32 side.

PROJECT
FX2LP - STM32F103 GPIF USB Bridge

SHEET TITLE
GPIF interconnect & handshake timing

DATE	REV	SHEET
2026-10-02	1.0	5 of 5